



CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT342-[Section] System Integration and Architecture

System Design Document (SDD)

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Version	Date	Author	Changes Made	Status
0.1	2/21/2026	Michelu Tia A. Lim	Initial draft	Draft
0.2	[Date]	[Your Name]	Added API specifications	Review
0.3	[Date]	[Your Name]	Updated database design	Review
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EXECUTIVE SUMMARY

1.1 Project Overview

ArtSpire is a web-based digital art publishing platform that enables users to upload, manage, and showcase their artwork securely. The system includes a Spring Boot backend REST API and a React web application, integrated to provide a seamless and secure experience for artists and administrators.

The platform implements authentication using JWT, role-based access control, Google OAuth login, file upload functionality, email notifications, and external API integration to enhance user engagement. ArtSpire focuses on secure artwork management and structured system architecture suitable for a modern web application.

1.2 Objectives

1. Develop a fully functional digital art publishing platform with secure authentication, role-based access control, and artwork management features. Implement a three-tier architecture using Spring Boot (backend), React (web), and Android (mobile)
2. Implement a layered architecture using Spring Boot (backend) and React (web frontend) to ensure separation of concerns and maintainability.
3. Create RESTful APIs to handle user authentication, artwork CRUD operations, file uploads, and system integrations.
4. Design a responsive and user-friendly interface that allows artists to upload and manage their artwork efficiently.
5. Integrate essential system features including Google OAuth login, external API consumption, email notifications, and secure file storage.
6. Deploy the backend and frontend in production-ready environments for final demonstration.

1.3 Scope

Included Features:

- User registration and authentication (email/password with BCrypt hashing)
- JWT-based authentication and protected routes
- Google OAuth login integration
- Role-Based Access Control (USER and ADMIN roles)

- Artwork management (Create, View, Update, Delete)
- Secure file upload for artwork images
- External public API integration (display inspiration content on dashboard)
- Email notifications (welcome email and artwork upload confirmation)
- Admin panel for user and artwork management
- Responsive web interface (React frontend)
- PostgreSQL database with relational schema (minimum 5 tables)

Excluded Features:

- Payment gateway integration (not applicable to publishing platform)
- Native Android mobile application
- Real-time chat or messaging system
- Artwork marketplace or auction system
- Push notifications
- Advanced analytics dashboard
- NFT or blockchain integration

1.0 INTRODUCTION

1.1 Purpose

This document serves as the comprehensive design specification for the ArtSpire system. It outlines the system requirements, architectural design, API structure, database schema, security configuration, and integration features necessary for implementation.

The purpose of this document is to guide development, ensure proper system architecture, and verify that all required components — including authentication, role-based access control, file upload, external API integration, Google OAuth, and email services — are correctly implemented and integrated into a secure and maintainable web application.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

Project Name: ArtSpire

Domain: Creative / Digital Art Publishing Platform

Primary Users:

1. Artist(Regular Users)
2. Administrators

Problem Statement: Many aspiring and independent artists need a secure platform to upload, organize, and showcase their artwork online. Existing platforms are often overloaded with commercial features, making it difficult for artists to focus on portfolio presentation and content control.

Solution: ArtSpire is a secure web-based digital art publishing platform that allows users to register, authenticate, upload artwork files, and manage their portfolios. The system implements JWT authentication, role-based access control, Google OAuth login, email notifications, and structured architecture to ensure secure and efficient artwork management.

2.2 Core User Journeys

Journey 1: First-Time Artist Registration & Artwork Upload

1. User visits the ArtSpire web application
2. Clicks "Sign Up" and creates an account (email/password or Google login)
3. Receives a welcome email confirmation
4. Logs into the system
5. Navigates to the "Upload Artwork" page
6. Uploads an artwork image file
7. Enters artwork details (title, description, category)
8. Submits the form and receives upload confirmation
9. Views the uploaded artwork in their profile

Journey 2: Returning Artist Managing Artwork

1. User logs in using email/password or Google OAuth
2. Accesses their dashboard

3. Views list of previously uploaded artworks
4. Edits artwork details (title/description)
5. Uploads a new artwork
6. Deletes an existing artwork if needed

Journey 3: Administrator Content Management

1. Admin logs in with admin credentials
2. Navigates to the admin dashboard
3. Views list of all users
4. Views all uploaded artworks
5. Deletes inappropriate artwork if necessary
6. Manages user roles (if applicable)

2.3 Feature List (MoSCoW)

MUST HAVE

1. User authentication (register, login, logout)
2. JWT-based authentication and protected routes
3. Role-Based Access Control (USER and ADMIN)
4. Artwork management (Create, View, Update, Delete)
5. Secure file upload and storage for artwork images
6. Google OAuth login integration
7. Email notifications (welcome email and artwork upload confirmation)
8. Admin panel (user and artwork management)
9. External API integration (display art inspiration content)

SHOULD HAVE

1. Artwork categories and filtering
2. Search functionality for artworks
3. Basic input validation feedback
4. Responsive design for different screen sizes
5. /me endpoint to retrieve authenticated user information

COULD HAVE

1. Artwork likes or simple engagement counter
2. Artist profile customization (bio, profile picture)
3. Basic dashboard statistics (total artworks uploaded)

WON'T HAVE

1. Payment gateway integration

2. Artwork marketplace or auction system
3. Real-time chat or messaging system
4. NFT or blockchain integration
5. Advanced analytics dashboard
6. Multi-language support

2.4 Detailed Feature Specifications

Feature: User Authentication

- **Screens:** Registration, Login, Forgot Password, Reset Password
- **Fields:** Email, Password, Confirm Password
- **Validation:** Valid email format, Password strength requirement, Email uniqueness during registration, Existing email validation for password reset, Password confirmation must match
- **API Endpoints:** POST /auth/register, POST /auth/login, POST /auth/logout, POST /auth/forgot-password, POST /auth/reset-password, GET /auth/me
- **Security:** JWT token generation and validation, Password hashing using Bcrypt, Protected routes for authenticated users, Reset token with expiration time, Role-based access control (USER, ADMIN)

Feature: Artwork Management (Core Business Module)

- **Screens:** Artwork Listing, Artwork Detail, Upload Artwork, Edit Artwork
- **Display:** Artwork image, Title, Description, Category, Artist (uploaded by), Upload date
- **Functions:** Create artwork (with file upload), View artwork, Update artwork, Delete artwork, Search artwork by title or category
- **API Endpoints:** GET /artworks, GET /artworks/{id}, GET /artworks/search, POST /artworks, PUT /artworks/{id}, DELETE /artworks/{id}
- **File Handling:** Image uploaded using multipart/form-data, File stored on server, File path saved in database, File linked to artwork record

Feature: Admin Panel

- **Screens:** Admin Dashboard, User Management, Artwork Management
- **Functions:** View all users, Change user roles, Delete inappropriate artworks, View all uploaded artworks
- **Access Control:** ADMIN role required, API-level role restriction, UI-level access restriction
- **API Endpoints:** GET /admin/users, PUT /admin/users/{id}/role, DELETE /admin/users/{id}, DELETE /admin/artworks/{id}

Feature: External API Integration

- **Purpose:** Display art inspiration or trending artworks from a public art API
- **Functionality:** Backend consumes external public API, Data processed and returned to frontend, Displayed on homepage or dashboard
- **API Endpoints:** GET /external/art-feed

Feature: Email Notification System

- **Screens:** No dedicated screen (Triggered by system events)
- **Functions:** Send welcome email, send password reset email, send artwork upload confirmation
- **Technology:** SMTP integration with real email delivery
- **Triggers:** After registration, after password reset request, after artwork upload

Feature: Google OAuth Login

- **Screens:** Login (Google Sign-In option)
- **Functions:** Authenticate user using Google account, auto-register new users, generate system JWT after successful login
- **Security:** OAuth2 authentication flow, JWT token generation, role assignment upon account creation
- **API Endpoints:** GET /oauth2/authorization/google, OAuth success handler endpoint

2.5 Acceptance Criteria

AC-1: Successful User Registration

- Given I am a new user
- When I enter a valid email and strong password
- And confirm password matches
- And click "Create Account"
- Then my account should be created
- And my password should be securely hashed
- And I should receive a welcome email
- And I should be automatically logged in
- And redirected to the homepage

AC-2: Artwork Upload

- Given I am logged in as a registered user
- When I upload an artwork with valid title, description, and image file
- And click "Submit"
- Then the artwork should be saved in the database

- And the image file should be stored on the server
- And I should receive an artwork upload confirmation email
- And the artwork should appear in the public artwork listing

AC-3: Forgot Password Flow

- Given I have an existing account
- When I enter my registered email in the Forgot Password page
- Then I should receive a password reset email
- When I submit a new valid password
- Then my password should be updated
- And the new password should be securely hashed
- And I should be able to log in using the new password

AC-4: Admin Artwork Management

- Given I am logged in as an administrator
- When I view the artwork management page
- Then I should see all uploaded artworks
- When I delete an inappropriate artwork
- Then it should be removed from the system
- And it should no longer appear in the public listing

AC-5: External API Integration

- Given I visit the homepage
- When the system requests data from the external art API
- Then external artwork data should be displayed
- And the data should load without breaking the system

3.0 NON-FUNCTIONAL REQUIREMENTS

3.1 Performance Requirements

- API response time: ≤ 2 seconds for 95% of requests
- Web page load time: ≤ 3 seconds on broadband connection
- Image upload processing time: ≤ 5 seconds for standard image sizes (≤ 5MB)
- Support at least 100 concurrent users
- Database queries complete within 500ms
- External API data retrieval should not exceed 3 seconds

3.2 Security Requirements

- 3.2 Security Requirements
- HTTPS for all communications
- JWT-based authentication
- Password hashing using BCrypt (salt rounds = 12)
- SQL injection prevention via ORM and parameterized queries
- XSS protection through input validation and output encoding
- Rate limiting: 100 requests per minute per IP
- Admin endpoints require role verification
- File upload validation (file type and size restrictions)
- Reset password tokens must expire after defined time

3.3 Compatibility Requirements

- **Web Browsers:** Chrome, Firefox, Safari, Edge (latest 2 versions)
- **Android:** API Level 24+ (Android 7.0+)
- **Screen Sizes:** Mobile (360px+), Tablet (768px+), Desktop (1024px+)
- **Operating Systems:** Windows 10+, macOS 10.15+, Linux Ubuntu 20.04+

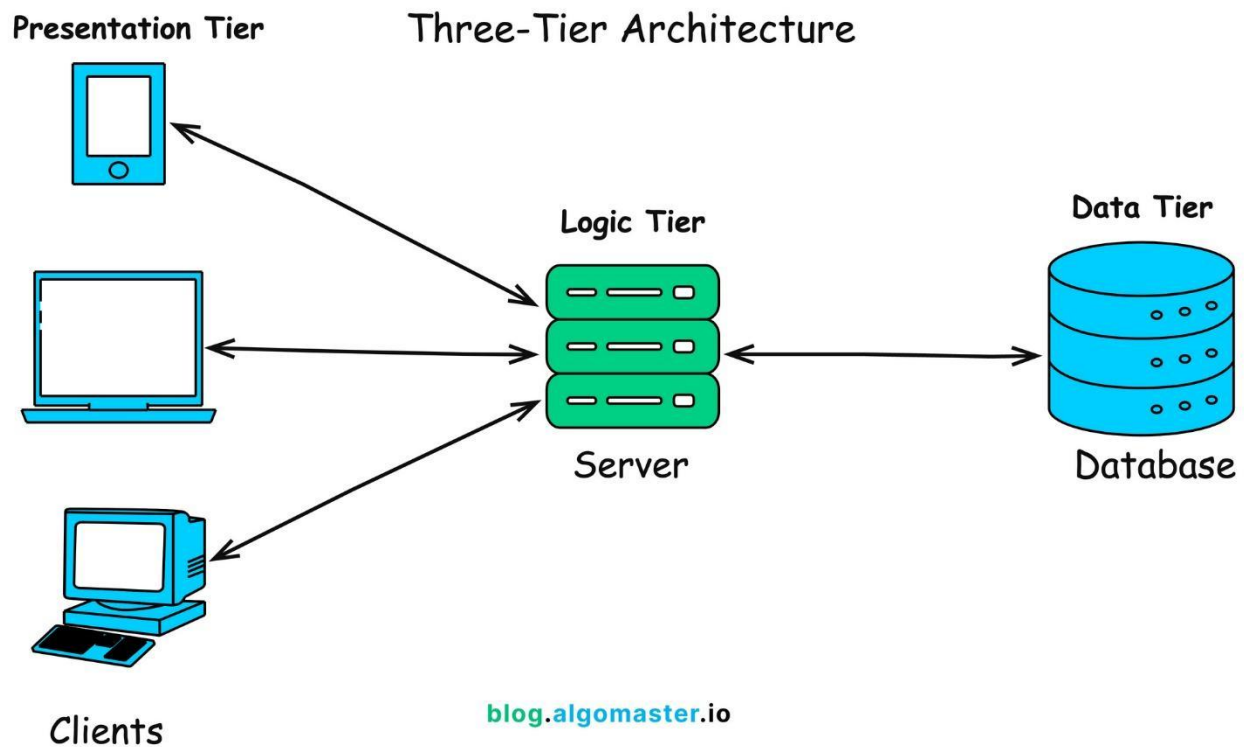
3.4 Usability Requirements

- New users should be able to register and upload their first artwork within 5 minutes
- Consistent navigation across all pages
- Clear and descriptive error messages with recovery guidance
- Responsive design across devices
- Touch targets minimum 44x44px on mobile
- Keyboard navigation support
- Simple and intuitive artwork upload process

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram

Note: This should be a component diagram



Technology Stack:

- **Backend:** Java 17, Spring Boot 3.x, Spring Security, Spring Data JPA
- **Database:** PostgreSQL 14+
- **Web Frontend:** React 18, TypeScript, Tailwind CSS, Axios
- **Mobile:** Kotlin, Jetpack Compose, Retrofit, Room
- **Build Tools:** Maven (Backend), npm/yarn (Web), Gradle (Android)
- **Deployment:** Railway/Heroku (Backend), Vercel/Netlify (Web), APK (Mobile)

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

Base URL	https://artspire-api.com/api/v1
Format	All requests and responses use JSON Content-Type: application/json File uploads use multipart/form-data
Authentication	Bearer token (JWT) in Authorization header
Response Structure	{ "success": boolean, "data": object null,

	<pre> "error": { "code": string, "message": string, "details": object null }, "timestamp": string } </pre>
--	--

5.2 Endpoint Specifications

Authentication Endpoints

User Registration

Description	User Registration
API URL	/auth/register
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre> { "email": <email>, "password": <password>, "firstname": <firstname>, "lastname": <lastname>, } </pre>
Response Structure	<pre> { "success": boolean, "data": { "user": { "email": "<email>", "firstname": "<firstname>", "lastname": "<lastname>" }, "accessToken": "<token>", "refreshToken": "<token>" }, "error": { "code": string, "message": string, "details": object null }, "timestamp": "string" } </pre>

User Login

Description	User Login
API URL	/auth/login
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre>{ "email": <email>, "password": <password>, }</pre>
Response Structure	<pre>{ "success": boolean, "data": { "user": { "email": "<email>", "firstname": "<firstname>", "lastname": "<lastname>", "role": "<role>" }, "accessToken": "<token>", "refreshToken": "<token>" }, "error": { "code": string, "message": string, "details": object null }, "timestamp": "string" }</pre>

5.3 Error Handling

HTTP Status Codes

- 200 OK - Successful request
- 201 Created - Resource created
- 400 Bad Request - Invalid input
- 401 Unauthorized - Authentication required/failed
- 403 Forbidden - Insufficient permissions
- 404 Not Found - Resource doesn't exist
- 409 Conflict - Duplicate resource
- 500 Internal Server Error - Server error

Error Code Examples

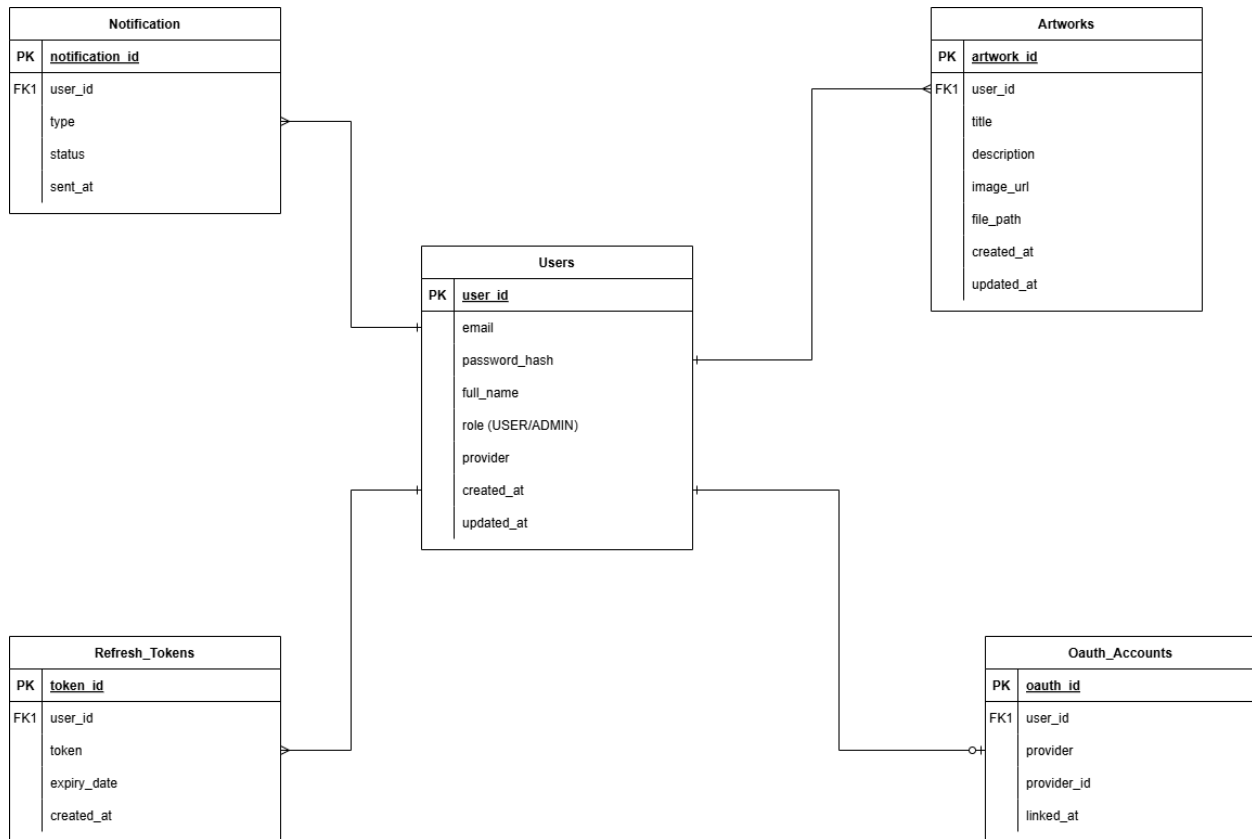
Example 1 – Invalid Credentials	<pre>{ "success": false, "data": null, "error": { "code": "AUTH-001", "message": "Invalid credentials", "details": "Email or password is incorrect" }, "timestamp": "2024-01-28T10:30:00Z" }</pre>
Example 2 – Validation Failed	<pre>{ "success": false, "data": null, "error": { "code": "VALID-001", "message": "Validation failed", "details": { "email": "Email is required", "password": "Must be at least 8 characters" } }, "timestamp": "2024-01-28T10:30:00Z" }</pre>

Common Error Codes

- AUTH-001: Invalid credentials
- AUTH-002: Token expired
- AUTH-003: Insufficient permissions
- VALID-001: Validation failed
- DB-001: Resource not found
- DB-002: Duplicate entry
- BUSINESS-001: Insufficient stock
- SYSTEM-001: Internal server error

6.0 DATABASE DESIGN

6.1 Entity Relationship Diagram



Detailed Relationships:

- **One-to-Many:** User → Artwork (A user can upload multiple artworks. Each artwork belongs to one user.)
- **One-to-Many:** User → RefreshTokens (A user can have multiple active refresh tokens. Each refresh token belongs to one user.)
- **One-to-Many:** User → Notifications (A user can receive multiple email notifications. Each notification belongs to one user.)
- **One-to-One (Optional):** User → OAuthAccount (A user may have one linked Google OAuth account. Each OAuth account is linked to one user.)

Key Tables:

1. **users** – Stores user account information and authentication credentials.
2. **artworks** – Stores uploaded artwork details and file references.
3. **refresh_tokens** – Stores JWT refresh tokens for session management.
4. **notifications** – Stores email notification logs (welcome email, artwork upload confirmation).

5. **oauth_accounts** – Stores Google OAuth account associations.

Table Structure Summary:

- **users:** id, email, password_hash, full_name, role, provider, created_at, updated_at
- **artworks:** id, title, description, image_url, file_path, user_id, created_at, updated_at
- **refresh_tokens:** id, user_id, token, expiry_date, created_at
- **notifications:** id, user_id, type, status, sent_at
- **oauth_accounts:** id, user_id, provider, provider_id, linked_at

7.0 UI/UX DESIGN

7.1 Web Application Wireframes

Registration Page

The wireframe shows a 'Create Account' form with the following elements:

- Create Account** (Section Header)
- Email** (Label) with a text input field and a light gray feedback bar below it.
- Password** (Label) with a text input field and a light gray feedback bar below it.
- Confirm Password** (Label) with a text input field and a light gray feedback bar below it.
- Password Strength:** (Label) with a horizontal progress bar.
- Register** (Dark blue button)
- Already have an account? Login** (Text link)

Login Page

Login

Email

Password

Login

Sign in with Google

Forgot Password?

Don't have an account? [Create Account](#)

Forgot Password Page

Forgot Password

Email

Send Reset Link

[Back to Login](#)

Reset Password Page

Reset Password

New Password

Confirm Password

Reset Password

Artwork Listing (Homepage / Dashboard)

LOGO ArtSpire

Search by title/category

Upload Artwork

Profile

Artwork Gallery

Tip: You can only edit/delete your own artworks

View

View

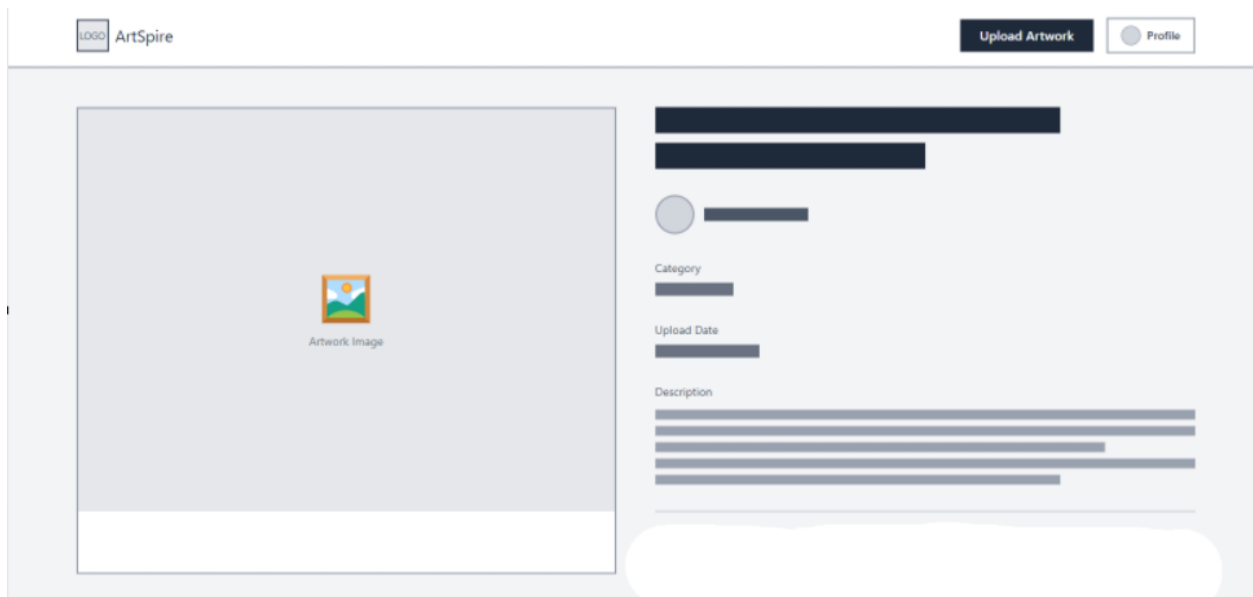
View

View

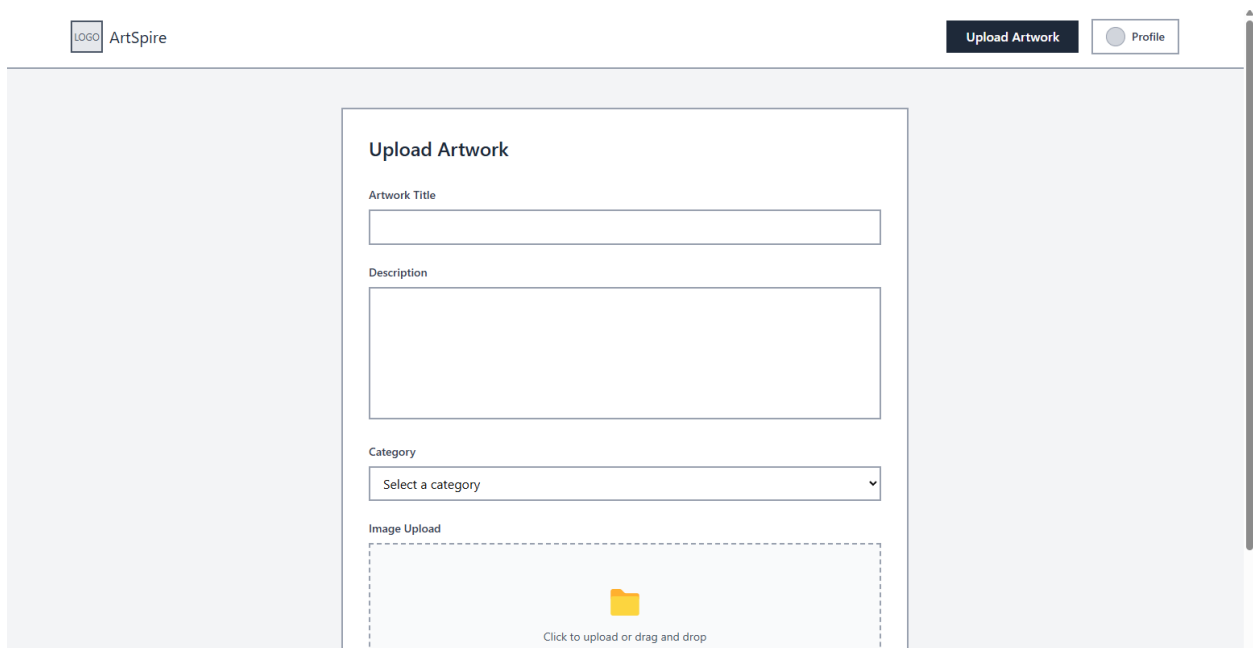
View

View

Artwork Detail Page



Upload Artwork Page



Edit Artwork Page

LOGO ArtSpire

Upload ArtworkProfile

Edit Artwork

Artwork Title

Description

This is a sample description for the artwork that has been pre-filled for editing purposes.

Category

Digital Painting

Image Upload

Admin Dashboard

LOGO ArtSpire

DashboardUser ManagementArtwork Management

Admin Dashboard

Total Users

Total Artworks

Recent Activity

User Management Page

LOGO ArtSpire

Dashboard

User Management

Artwork Management

User Management

Email	Full Name	Role	Actions
		User	Delete
		User	Delete
		User	Delete
		User	Delete
		User	Delete
		User	Delete
		User	Delete
		User	Delete
		User	Delete
		User	Delete
		User	Delete
		User	Delete

Artwork Management Page

LOGO ArtSpire

Dashboard

User Management

Artwork Management

Artwork Management

Delete

Delete

Delete

22

7.2 Mobile Application Wireframes

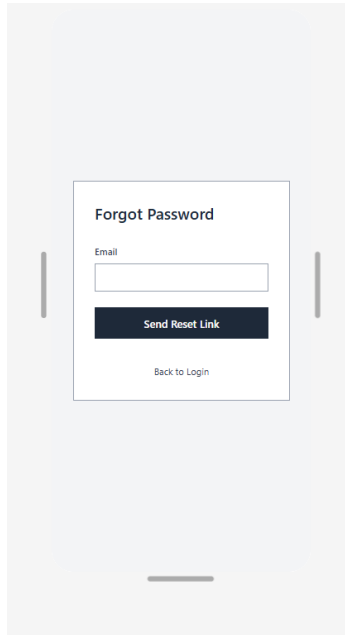
Registration Page

The registration page wireframe is displayed on a mobile device screen. It features a white card with a light gray border. At the top of the card is the title "Create Account". Below the title are four input fields, each with a label above it: "Email", "Password", "Confirm Password", and "Password Strength:". Each input field has a light gray placeholder bar below it. Below the input fields is a dark blue button with the text "Register" in white. At the bottom of the card, there is a link that says "Already have an account? Login". The card is centered on a light gray background that represents the mobile device screen, with a thin horizontal line at the bottom indicating the home indicator bar.

Login Page

The login page wireframe is displayed on a mobile device screen. It features a white card with a light gray border. At the top of the card is the title "Login". Below the title are two input fields, each with a label above it: "Email" and "Password". Each input field has a light gray placeholder bar below it. Below the input fields is a dark blue button with the text "Login" in white. Below the "Login" button is a button with the text "Sign in with Google". At the bottom of the card, there is a link that says "Forgot Password?" and a link that says "Don't have an account? Create Account". The card is centered on a light gray background that represents the mobile device screen, with a thin horizontal line at the bottom indicating the home indicator bar.

Forgot Password Page



A mobile app UI mockup for the 'Forgot Password' page. The page has a light gray background. In the center, there is a white card with a thin gray border. At the top of the card, the title 'Forgot Password' is displayed in a bold, dark gray font. Below the title, the label 'Email' is positioned above a white text input field. Underneath the input field is a dark blue button with the text 'Send Reset Link' in white. At the bottom of the card, the text 'Back to Login' is displayed in a smaller, dark gray font. The entire card is flanked by two vertical gray bars, and a horizontal gray bar is visible at the very bottom of the screen, representing a mobile device's home indicator.

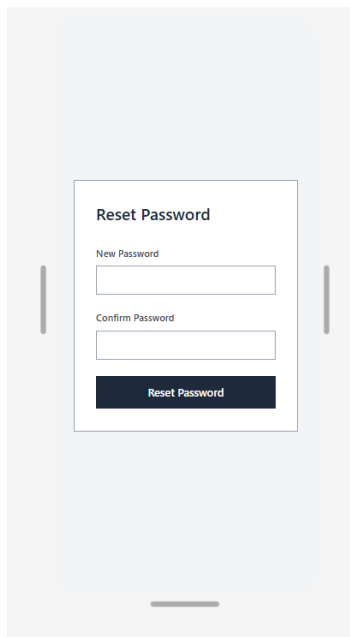
Forgot Password

Email

Send Reset Link

Back to Login

Reset Password Page



A mobile app UI mockup for the 'Reset Password' page. The page has a light gray background. In the center, there is a white card with a thin gray border. At the top of the card, the title 'Reset Password' is displayed in a bold, dark gray font. Below the title, the label 'New Password' is positioned above a white text input field. Underneath the first input field, the label 'Confirm Password' is positioned above a second white text input field. Below the second input field is a dark blue button with the text 'Reset Password' in white. The entire card is flanked by two vertical gray bars, and a horizontal gray bar is visible at the very bottom of the screen, representing a mobile device's home indicator.

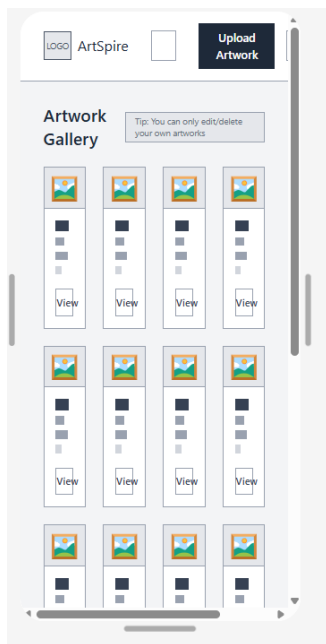
Reset Password

New Password

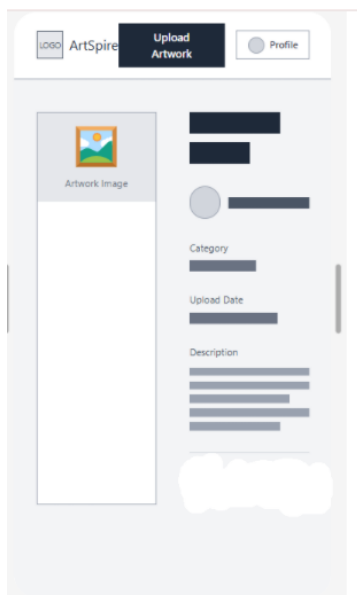
Confirm Password

Reset Password

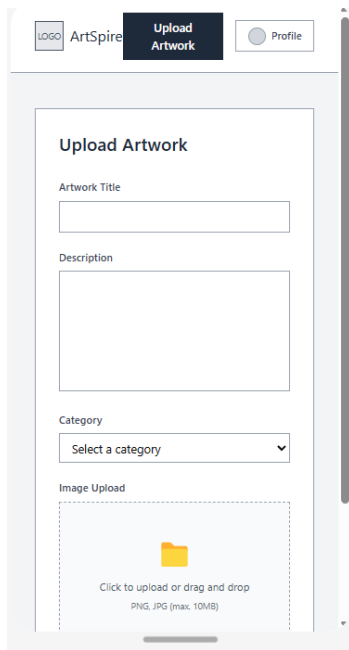
Artwork Listing (Homepage / Dashboard)



Artwork Detail Page



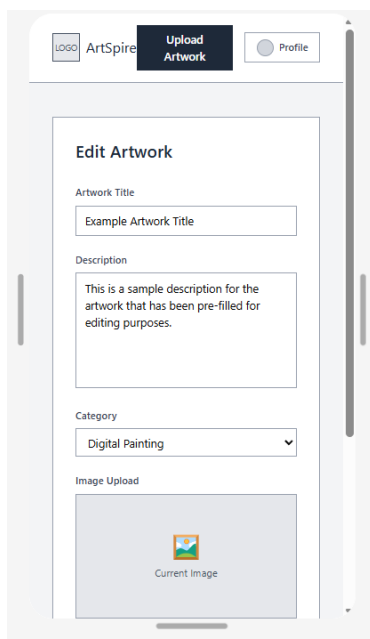
Upload Artwork Page



Mobile app mockup of the Upload Artwork page. The header bar contains a logo, the text 'ArtSpire', a dark 'Upload Artwork' button, and a 'Profile' button. The main content area is titled 'Upload Artwork' and includes the following fields:

- Artwork Title:** A text input field.
- Description:** A large text area.
- Category:** A dropdown menu with the placeholder text 'Select a category'.
- Image Upload:** A dashed border box containing a yellow folder icon and the text: 'Click to upload or drag and drop' and 'PNG, JPG (max. 10MB)'.

Edit Artwork Page



Mobile app mockup of the Edit Artwork page. The header bar is identical to the upload page. The main content area is titled 'Edit Artwork' and includes the following fields:

- Artwork Title:** A text input field containing the placeholder text 'Example Artwork Title'.
- Description:** A large text area containing the placeholder text: 'This is a sample description for the artwork that has been pre-filled for editing purposes.'
- Category:** A dropdown menu with 'Digital Painting' selected.
- Image Upload:** A box containing a small image icon and the text 'Current Image'.

Mobile-Specific Features:

- Touch-optimized buttons (min 44x44px)
- Gesture support (swipe, pull-to-refresh)
- Offline caching for product images
- Bottom navigation for main actions
- Simplified forms for mobile input

Design System:

- **Colors:** Primary (#111827), Secondary (#6B7280), Success (#10B981), Error (#EF4444)
- **Typography:** Inter font family, responsive sizing
- **Spacing:** 8px grid system
- **Components:** Consistent buttons, inputs, cards, modals
- **Responsive:** Mobile-first approach, breakpoints at 640px, 768px, 1024px

8.0 PLAN

8.1 Project Timeline

Phase 1: Planning & Design (Week 1-2)

Week 1: Requirements & Architecture

Day 1-2: Project setup and documentation

Day 3-4: Complete FRS and NFR

Day 5-7: System architecture design

Week 2: Detailed Design

Day 1-2: API specification

Day 3-4: Database design

Day 5-6: UI/UX wireframes

Day 7: Implementation plan finalization

Phase 2: Backend Development (Week 3-4)

Week 3: Foundation

Day 1: Spring Boot setup with dependencies

Day 2: Database configuration and entities

Day 3: JWT authentication implementation

Day 4: User management endpoints

Day 5: Product CRUD operations

Week 4: Core Features

Day 1: Cart functionality

Day 2: Order management

Day 3: Search and filtering

Day 4: Error handling and validation

Day 5: API documentation and testing

Phase 3: Web Application (Week 5-6)

Week 5: Frontend Foundation

Day 1: React setup with TypeScript

Day 2: Authentication pages (login, register)

Day 3: Product listing page

Day 4: Product detail page

Day 5: Shopping cart implementation

Week 6: Complete Web Features

Day 1: Checkout flow

Day 2: Order history and confirmation

Day 3: Admin dashboard

Day 4: Responsive design polish

Day 5: API integration and testing

Phase 4: Mobile Application (Week 7-8)

Week 7: Android Foundation

Day 1: Android Studio setup and project structure

Day 2: Authentication screens

Day 3: Product browsing

Day 4: Shopping cart

Day 5: API service layer

Week 8: Complete Mobile App

Day 1: Checkout flow

Day 2: Order management

Day 3: UI polish and animations

Day 4: Testing on emulator/device

Day 5: APK generation and documentation

Phase 5: Integration & Deployment (Week 9-10)

Week 9: Integration Testing

Day 1: End-to-end testing across platforms

Day 2: Bug fixes and optimization

Day 3: Security review

Day 4: Performance testing

Day 5: Documentation updates

Week 10: Deployment

Day 1: Backend deployment (Railway/Heroku)

Day 2: Web app deployment (Vercel/Netlify)

Day 3: Mobile APK distribution

Day 4: Final testing

Day 5: Project submission

Milestones:

- **M1 (End Week 2):** All design documents complete
- **M2 (End Week 4):** Backend API fully functional
- **M3 (End Week 6):** Web application complete
- **M4 (End Week 8):** Mobile application complete
- **M5 (End Week 10):** Full system deployed and integrated

Critical Path:

1. Authentication system (Week 3)
2. Product catalog API (Week 3-4)
3. Shopping cart functionality (Week 4)
4. Checkout process (Week 6)
5. Cross-platform testing (Week 9)

Risk Mitigation:

- Start with simplest working version of each feature
- Test integration points early and often
- Keep backup of working versions
- Focus on core functionality before enhancements